# Unit 1 Lecture 1 Chapter 5 Problems

Monday, August 26, 2024 6:01 PM

Unit 1 Lecture 1 Chapter 5 Problems

# Exercise 5.2

① $i_{DSAT} = \frac{k_n V_{ov}^2}{2}$, ② $k_n' = \mu_n C_{ox}$ ③ $C_{ox} = \frac{\epsilon_{ox}}{t_{ox}}$, ④ $k_n = \frac{W}{L} k_n'$

If $t_{ox}$ = 4 nm and $\mu_n$ = 450 cm²/Vs find $C_{ox}$, $k_n'$ and the overdrive voltage required to operate a transistor having W/L =20 in saturation with $i_D$ = 0.3 mA. What is the minimum $v_{DS}$ needed?

(a) $C_{ox} = \frac{\epsilon_{ox}}{t_{ox}} = \frac{3.45 \times 10^{-8} F/cm}{4 \times 10^{-7} cm} = \boxed{8.625 \times 10^{-7} F/cm^2}$ $C_{ox}$

(b) $k_n' = \mu_n C_{ox} = (450 \, cm^2/Vs)(8.625 \times 10^{-7} F/cm^2) = \boxed{388 \, \mu A/V^2}$ $k_n'$

$k_n = \left(\frac{W}{L}\right)(k_n') = (20)(388 \, \mu A/V^2) = \boxed{7.8 \, mA/V^2}$

$V_{ov} = V_{DSAT}$ @ $I_{DSAT}$

(c) $i_{DSAT} = \frac{k_n V_{ov}^2}{2} \Rightarrow V_{ov} = \sqrt{\frac{2 I_{DSAT}}{k_n}} = \sqrt{\frac{2 \times 0.3 \, mA}{7.8 \, mA/V^2}}$

$\boxed{V_{DSAT} = 0.28}$ $\boxed{V_{ov} = 0.28 V}$

# Problem 5.4

(5) $\boxed{r_{DS} = \frac{1}{k_n V_{OV}}}$ (6) $V_{GS} - V_T = V_{OV}$

An NMOS transistor is operated with a small $v_{DS}$. By what factor will $r_{DS}$ change in the following situations?

a. $V_{OV}$ is doubled → $r_{DSD} = \frac{1}{2} r_{DSO}$ $\quad r_{DSD} = \frac{1}{k_n (2) V_{OV}} = \frac{1}{2} r_{DSO}$

b. The device width is doubled → $r_{DSD} = \frac{1}{2} r_{DSO} \Rightarrow r_{DS} = \frac{1}{k_n V_{OV}} = \frac{1}{k_n' \left(\frac{W}{L}\right) V_{OV}}$ 2x

c. The device length is doubled

d. The oxide thickness is halved $\quad r_{DSD} = 2 r_{DS}$

$$r_{DS} = \frac{1}{k_n' \left(\frac{W}{L}\right) V_{OV}} = \frac{1}{\mu C_{ox} \left(\frac{W}{L}\right) V_{OV}} = \frac{1}{\mu \frac{\varepsilon_{ox}}{t_{ox}} \left(\frac{W}{L}\right) V_{OV}} \Rightarrow r_{DS} \Rightarrow \frac{1}{2}$$

smaller

# Problem 5.6

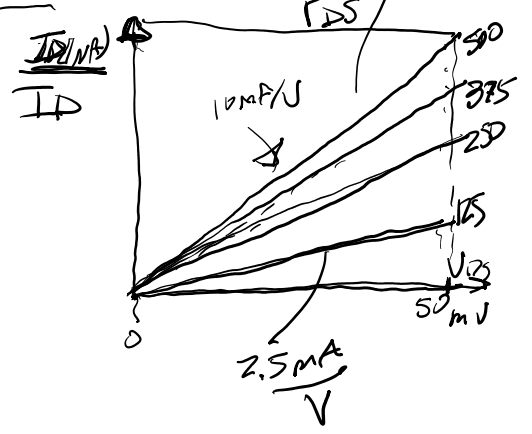
$g_{DS} = k_n V_{ov}$ $\quad r_{DS} = \frac{1}{g_{DS}}$ $\quad V_{ov} = V_{GS} - V_t$

An NMOS transistor has $k_n = 5mA/V^2$ and $V_{tn} = 0.5V$. Sketch the $i_D$-$v_{DS}$ characteristic for $v_{DS}$ ranging from 0 to 50 mV when $v_{GS}$ = 0.5V, 1.0V, 1.5V, 2.0V, and 2.5V. Give the value of $r_{DS}$ for each value of $v_{GS}$.

| $V_{GS}$ | $V_{ov}$ | $g_{DS}$ | $i_D$ ($V_{DS}$ = 50mV) | $r_{DS}$ |
|---|---|---|---|---|
| → 0.5 | 0 | 0 | 0 | ∞ CUTOFF |
| 1.0 | 0.5 | 2.5 mA/V | 125 μA | 400 |
| 1.5 | 1.0 | 5.0 mA/V | 250 μA | 200 |
| 2.0 | 1.5 | 7.5 mA/V | 375 μA | 133 |
| 2.5 | 2.0 | 10 mA/V | 500 μA | 100 |

$i_D = g_{DS} V_{DS}$

$= \frac{V_{DS}}{r_{DS}}$ linear




Oxford University Publishing
Microelectronic Circuits by Adel S. Sedra and Kenneth C. Smith (0195323033)

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